

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. We construct a projectively consistent positive phase label on the original microscopic sample space from isolated rate components and an explicit boundary band. Component rates are basin-interior lower-semicontinuous recovery costs, so boundary points are not assigned the unrestricted rate by fiat. Positive component operators provide phase-local response charts. Exact nonlinear aggregation is log-sum-exp at finite volume and a maximum with an active-phase subdifferential in the limit, never a linear mixture of log values. Target-dependent shell costs, Gaussian mixtures, and labelled law-level commutation give a standalone positive phase-sheaf theorem without interpreting spectral projections as probabilities.

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1. ROUND-FOURTEEN POSITIVE CLOSURE: A POSITIVE PHASE SHEAF AND LOGARITHMIC NONLINEAR MIXTURES

The phase label is a positive measurable coordinate on the original microscopic sample space. Component rates are lower-semicontinuous basin-internal recovery costs, not the unrestricted rate on a basin boundary. Non-linear log-Laplace values combine by an exact log-sum-exp formula, never by a linear mixture.

1.1. Canonical rate basins. Let X_n be a platform path variable with good rate I , and let $M : \mathcal{X} \rightarrow (\mathcal{O}, d_{\mathcal{O}})$ be the platform order parameter fixed in A3 or B2. On a regular finite-phase chart assume the zero/low-rate set of I_M has finitely many isolated connected components $K_1, \dots, K_J$. Choose δ below one quarter of their mutual distance and define the intrinsic Voronoi basins

$$G_j = \{x : d(Mx, K_j) < \delta, d(Mx, K_j) < d(Mx, K_\ell), \ell \neq j\},$$

and $G_\partial = \mathcal{X} \setminus \bigcup_j G_j$. Ties are assigned to the boundary label. This construction depends only on the metric rate components and is invariant under isometric reparametrization of the order parameter.

Let $J_n = J(X_n) \in \{1, \dots, J, \partial\}$.

Theorem 1.1 (Exact positive phase sheaf). *For every n ,*

$$\mathbb{P}_n = \sum_{j \in \{1, \dots, J, \partial\}} w_{n,j} \mathbb{P}_{n,j}, \quad w_{n,j} = \mathbb{P}_n(J_n = j),$$

where every $\mathbb{P}_{n,j}$ is a probability conditional on the same measurable label. The label is unchanged under every finite observable projection, so it defines one projective labelled process. The boundary component is retained at its actual cost.

Proof. The intrinsic Voronoi rule is Borel and gives a disjoint partition. The identity is the law of total probability. Since the label is defined once from the full order parameter, all projected laws carry the same random label; no projection-dependent relabelling occurs. $\square$

1.2. Component rates with boundary recovery. Define the basin-internal regularization

$$I_j(x) = \liminf_{\epsilon \downarrow 0} \{I(y) - \alpha_j : y \in G_j, d(x, y) < \epsilon\}, \quad \alpha_j = \inf_{y \in G_j} I(y),$$

with value $+\infty$ if the set is empty. Define I_∂ in the same way.

Lemma 1.2 (Basin-internal recovery). *For the A3 and B2 platform rates, every x with $I_j(x) < \infty$ has a microscopic recovery sequence whose empirical state remains in G_j and whose cost tends to $\alpha_j + I_j(x)$.*

Proof. By definition choose $y_m \in G_j$ tending to x with $I(y_m) \rightarrow \alpha_j + I_j(x)$. A3 recovers each finite-rate controlled renewal law by an actual

stationary controlled kernel and legal connectors. B2 recovers each finite-action density/contact path by a smooth positive exposed pair and the exact kinetic Hodge repair. Because G_j is open and M is continuous, the recovery may be chosen in a sufficiently small neighborhood of y_m , hence inside G_j . Diagonalizing in m gives the sequence. The same argument applies to the boundary component using its own interior in the labelled topology. $\square$

Theorem 1.3 (Positive component LDP). *If $-n^{-1} \log w_{n,j} \rightarrow \alpha_j$, the conditional laws $\mathbb{P}_{n,j}$ satisfy good LDPs with rates I_j . The joint labelled law has rate*

$$\mathcal{J}(j, x) = \alpha_j + I_j(x),$$

and forgetting the label gives

$$I(x) = \min_j \{\alpha_j + I_j(x)\}.$$

At a basin boundary, each component pays its own basin-internal recovery cost; it is not automatically $I(x) - \alpha_j$.

Proof. For a closed set use the platform upper bound on its intersection with the closed basin closure. For an open set and a finite-rate point, use Lemma 1.2. Division by $w_{n,j}$ subtracts α_j . Compactness follows because component sublevels are closed subsets of a compact I -sublevel. The finite-label joint LDP and the contraction principle give the final formula. $\square$

1.3. Component source charts. Let

$$\tilde{Q}_{n,j}(F) = \frac{1}{n} \log \mathbb{E}_n[e^{nF(X_n)} \mid J_n = j].$$

The unnormalized restricted pressure is

$$Q_{n,j}^{\text{un}}(F) = \frac{1}{n} \log \mathbb{E}_n[e^{nF(X_n)}; J_n = j] = -\frac{1}{n} \log w_{n,j} + \tilde{Q}_{n,j}(F).$$

Theorem 1.4 (Phase-conditioned analytic chart). *If the tilted minimizer of $\alpha_j + I_j - \nu(F)$ is unique and lies a positive rate distance from the labelled boundary, then $\tilde{Q}_{n,j}$ has a zero-free complex neighborhood with locally uniform first two derivatives. Its limiting derivatives are the component mean and covariance. The boundary label has the same theorem whenever it has an interior exposed phase; it is otherwise retained through its real Laplace principle without being declared negligible.*

Proof. For A3, impose the basin label on the controlled Markov-renewal kernel. The positive killed transfer operator has leakage to the other labelled components of size e^{-gn} , where g is the rate separation. Its Perron root is isolated on the component graph, and analytic perturbation in the source gives a zero-free coefficient chart.

For B2, the component condition is an initial/order-parameter constraint. B1 exact-number coefficient extraction and the source-dependent saddle apply with the positive basin indicator inserted. The interior rate gap makes the complement exponentially smaller relative to the component

coefficient. In both cases Rouché's theorem gives the finite-volume zero-free neighborhood and Cauchy estimates give derivative convergence. No real probability gap alone is used without the corresponding positive component operator/coefficient construction. $\square$

1.4. Exact nonlinear phase aggregation. Let $V_{n,j}(\Phi)$ be any component log-Laplace value, including a dynamic component Nisio/Doob value. Then exactly

$$V_n(\Phi) = \frac{1}{n} \log \sum_j \exp\{n[-\alpha_{n,j} + V_{n,j}(\Phi)]\}, \quad \alpha_{n,j} = -\frac{1}{n} \log w_{n,j}.$$

Theorem 1.5 (Logarithmic phase-mixture calculus). *If the component values converge locally uniformly, then*

$$V(\Phi) = \max_j \{-\alpha_j + V_j(\Phi)\}.$$

At a unique maximizer, derivatives equal that component's derivatives. At a tie, the subdifferential is the convex hull of the active component derivatives, with finite-volume subleading weights selecting subsequential mixtures. Linear evolution acts on component measures before the logarithm; the nonlinear values themselves are never averaged linearly.

Proof. The finite formula is the exact positive disintegration from Theorem 1.1. The elementary Laplace principle for a finite sum gives the maximum. Differentiate the finite log-sum-exp: the derivative is the softmax-weighted component derivative. At a unique maximizer the weight converges to one; at a tie every cluster point lies in the stated convex hull. This proves the nonlinear aggregation rule. $\square$

For dynamic values, retain the finite phase weights as a state coordinate. Component semigroups evolve each positive component, and Bayes/log-sum-exp updates the weights. This gives an augmented nonlinear semigroup even at coexistence.

1.5. Phase-conditioned shells and tangents. Let $C(X_n)$ be a conditioned macro variable. Suppose each active component has target rate

$$I_{j,C}(a) = \inf_{Cx=a} I_j(x)$$

and local coefficient $c_{n,j}(a)$.

Theorem 1.6 (Exact target-dependent shell mixture). *The phase weight after conditioning on a regular shell about a is proportional to*

$$w_{n,j} c_{n,j}(a) e^{-nI_{j,C}(a)}.$$

The boundary label is included with its own target rate and coefficient. A uniquely smallest exponent selects one phase; at a tie, the normalized subexponential coefficients determine subsequential mixtures. The Gaussian tangent in a selected phase is its component covariance, while at a tie the labelled

tangent is a finite Gaussian mixture rather than one artificially averaged Gaussian.

Proof. Intersect the exact positive disintegration with the shell. Apply the A2 or B1 component coefficient theorem to each positive term. Each term has its own constrained infimum $I_{j,C}(a)$, including basin-boundary recovery. Divide by their sum. Conditional Gaussian limits pass termwise because the number of labels is finite. $\square$

1.6. Commutation on the labelled law.

Theorem 1.7 (Positive labelled commutation theorem). *On the joint labelled microscopic law, measurable contraction, exact shell conditioning, component Gaussian scaling, and linear law evolution commute with positive phase disintegration. After applying the logarithm, their nonlinear aggregation is precisely Theorem 1.5; no linear-mixture commutation is asserted.*

Proof. Before the logarithm, all operations act on the finite positive measures $w_{n,j}\mathbb{P}_{n,j}$: pushforward, restriction, rescaling, and linear law evolution distribute over a finite sum. Conditioning renormalizes the resulting positive terms. Taking the logarithm after summation gives the exact log-sum-exp identity and no other aggregation rule. $\square$

Theorem 1.8 (Round-fourteen D1 closure). *Every regular finite-phase chart of the proved Sinai or hard-sphere platform has a projectively consistent positive phase sheaf with basin-internal component rates, component response charts, exact target-dependent shell weights, Gaussian mixtures, and logarithmic nonlinear dynamic aggregation. The phase cost is counted once and no spectral projection is interpreted as a probability.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.5, 1.6, and 1.7. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-fourteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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